=== UNIT 5: NUCLEIC ACID STRUCTURE, DNA REPLICATION & TRANSCRIPTION ===

>> 5.1 Nucleic Acid Structure and Polymorphism

Deoxyribonucleic acid (DNA) is the molecular repository of genetic instructions. The primary structure consists of linear polydeoxyribonucleotide chains formed by 3',5'-phosphodiester linkages between deoxyribose sugars and purine (Adenine [A], Guanine [G]) or pyrimidine (Cytosine [C], Thymine [T]) nitrogenous bases.

* Chargaff's Rules: In duplex DNA, [A] = [T] and [G] = [C]; purines equal pyrimidines ([A+G] = [T+C]); molar ratio [A+T]/[G+C] varies among species.
* Watson-Crick B-DNA Model: Right-handed double helix with antiparallel strands (5'->3' paired with 3'<-5'). Helical diameter is 2.0 nm (20 A). Helical pitch is 3.4 nm (34 A) containing ~10.5 base pairs per helical turn, with an axial rise of 0.34 nm (3.4 A) per base pair. Features a wide, deep Major Groove (1.2 nm) and a narrow, shallow Minor Groove (0.6 nm). Major groove displays distinct patterns of hydrogen bond donors, acceptors, and methyl groups that enable sequence-specific transcription factor binding without unwinding the duplex.
* Conformational Variants (B-DNA vs. A-DNA vs. Z-DNA):
- B-DNA: Right-handed, C2'-endo sugar pucker, anti glycosidic bond conformation, stable under physiological hydration (>92% relative humidity).
- A-DNA: Right-handed, compact 2.6 nm diameter, 11 bp/turn, C3'-endo sugar pucker, anti conformation. Adopted by DNA-RNA hybrids and double-stranded RNA.
- Z-DNA: Left-handed zig-zag backbone, 1.8 nm diameter, 12 bp/turn, alternating purine-pyrimidine sequence (e.g., poly d(GC)). Alternating sugar puckers: C2'-endo for pyrimidines (anti conformation) and C3'-endo for purines (syn conformation).
* Thermal Denaturation and Hyperchromicity: Heating duplex DNA disrupts inter-base hydrogen bonds and base stacking interactions. Denaturation causes a 30-40% increase in optical absorbance at 260 nm (Hyperchromic Shift). Melting temperature (Tm) is the midpoint of the thermal transition where 50% of the duplex is denatured. Tm increases linearly with G+C content (3 H-bonds vs. 2 H-bonds in A-T pairs) and ionic strength (cations shield negative phosphate backbone charges).

>> 5.2 Enzymology and Mechanism of DNA Replication

* Semi-Conservative Mechanism: Demonstrated by Matthew Meselson and Franklin Stahl (1958) using 15N/14N isotope density gradient ultracentrifugation in CsCl. Each daughter duplex contains one intact conserved parental template strand and one newly synthesized nascent strand.
* Prokaryotic Replication Machinery (E. coli):
- Origin of Replication (oriC): 245 bp locus containing three 13-mer A-T rich tandem repeats (DNA unwinding element) and five 9-mer DnaA binding consensus boxes.
- Initiation: DnaA-ATP oligomerizes at 9-mer boxes, wrapping DNA around a protein

filament. This introduces positive superhelical strain that drives open-complex melting of the 13-mer A-T rich unwinding element. DnaC chaperone loads DnaB hexameric helicase onto each single strand in opposite directions.
- Unwinding and Protection: DnaB unwinds duplex DNA 5'->3' consuming ATP. Single-Stranded DNA-Binding protein (SSB) coats exposed single strands in a cooperative manner to prevent re-annealing and secondary hairpin formation. DNA Topoisomerase II (DNA Gyrase: GyrA/GyrB) introduces negative supercoils ahead of the advancing replication fork to relieve positive superhelical torsional stress.
- Priming: DnaG Primase synthesizes short complementary RNA primers (10-12 nucleotides) providing a free 3'-OH group required by DNA polymerases.
- Elongation: DNA Polymerase III Holoenzyme (10-subunit asymmetric dimer). Catalytic core comprises alpha (polymerase), epsilon (3'->5' proofreading exonuclease), and theta (assembly stabilizer). The beta-sliding clamp (homodimer forming a 35 A central hole) encircles duplex DNA, conferring extraordinary processivity (>1,000 nucleotides/sec without dissociation). Loaded by the gamma-complex clamp loader.
- Semi-Discontinuous Synthesis: Leading strand synthesized continuously toward the replication fork. Lagging strand synthesized discontinuously away from the fork in short Okazaki fragments (1,000-2,000 nt in prokaryotes, 100-200 nt in eukaryotes).
- Primer Removal and Ligation: DNA Polymerase I utilizes its 5'->3' exonuclease nick-translation domain to degrade the RNA primer while simultaneously polymerizing DNA from the adjacent 3'-OH. DNA Ligase couples NAD+ (in bacteria) or ATP (in eukaryotes) hydrolysis to synthesize phosphodiester bonds, sealing the remaining single-strand nicks.

>> 5.3 Prokaryotic Transcription: Initiation, Elongation & Termination

* RNA Polymerase Holoenzyme: Bacterial core enzyme (alpha2-beta-beta'-omega, 400 kDa) carries catalytic polymerase activity but cannot recognize promoter sequences. Association with a Sigma factor (e.g., sigma-70 for housekeeping genes) forms the Holoenzyme (alpha2-beta-beta'-omega-sigma).
* Bacterial Promoters: Feature two conserved hexanucleotide consensus motifs positioned upstream of the +1 transcription start site: the -10 element (Pribnow Box, 5'-TATAAT-3') and the -35 element (5'-TTGACA-3'), separated by an optimal spacer of 17 +/- 1 base pairs.
* Transcription Stages:
1. Closed Complex: Sigma region 4.2 binds the -35 element; region 2.4 binds the -10 element.
2. Open Complex: Holoenzyme induces spontaneous unwinding of ~14 bp from -11 to +3 without ATP requirement.
3. Abortive Initiation: Enzyme repeatedly synthesizes and releases short transcripts (2-9 nt) without moving.
4. Promoter Clearance and Elongation: Once transcript length reaches ~10 nt, sigma factor is released. Core enzyme translocates along template processively at 40-50 nt/sec.

5. Termination: Two distinct modes:
- Rho-Independent (Intrinsic): Nascent transcript contains an inverted G-C rich repeat followed by a run of 6-8 uracils. The RNA folds into a stable hairpin loop that causes RNA Pol to pause. Weak rU-dA base pairs dissociate, releasing the transcript.
- Rho-Dependent: Hexameric ATP-dependent Rho helicase binds the C-rich, G-poor rut (Rho utilization) site on naked nascent RNA. Rho translocates 5'->3' along the transcript, catches the paused polymerase, and unwinds the RNA-DNA hybrid in the transcription bubble.

[EXAMPLE 1: Spectrophotometric Hyperchromicity and Tm Calculation]

Problem Context: A 1000 bp double-stranded DNA fragment has a base composition of 58% G+C. Calculate its theoretical melting temperature (Tm) in a physiological buffer containing 0.15 M Na+ using the Marmur-Doty empirical equation: Tm = 81.5 + 16.6 * log10([Na+]) + 0.41 * (%G+C) - (500 / Length). Explain why thermal denaturation increases A260.
Step 1 (Calculation of Ionic Strength Term):
log10([Na+]) = log10(0.15) = -0.8239.
16.6 * (-0.8239) = -13.68 deg C.
Step 2 (Calculation of G+C Contribution):
0.41 * 58 = +23.78 deg C.
Step 3 (Calculation of Length Correction):
500 / 1000 = 0.50 deg C.
Step 4 (Total Tm Determination):
Tm = 81.5 - 13.68 + 23.78 - 0.50 = 91.10 deg C.
Physical Basis of Hyperchromicity: In duplex DNA, aromatic purine and pyrimidine ring systems are stacked parallel to each other at 3.4 A intervals. Overlapping pi-electron orbitals induce strong dipole-dipole interactions that suppress UV light absorption (hypochromism). Upon thermal denaturation into single strands, base stacking is disrupted, exposing unconstrained conjugated ring systems to UV photons and causing a 35-40% increase in optical absorbance at 260 nm.

[EXAMPLE 2: Replisome Okazaki Fragment Synthesis Timing]

Problem Context: The E. coli genome consists of 4.64 * 10^6 base pairs, replicated bidirectionally from oriC by two replisomes in 40 minutes at 37 deg C. Calculate the replication fork movement velocity (bp/sec), the rate of Okazaki fragment initiation on the lagging strand, and the number of Okazaki fragments generated per replication cycle assuming an average Okazaki fragment length of 1,500 bp.
Step 1 (Replication Fork Velocity):
- Bidirectional replication means each of the 2 forks replicates half the genome:
4.64 * 10^6 bp / 2 = 2.32 * 10^6 bp.

- Total replication time = 40 min = 40 * 60 = 2,400 seconds.
- Fork velocity = 2.32 * 10^6 bp / 2,400 sec = 966.7 bp/second per fork (~1,000 bp/sec).

Step 2 (Lagging Strand Okazaki Fragment Cycle Time):
- Each Okazaki fragment = 1,500 bp.
- Time required to synthesize 1 fragment = 1,500 bp / 966.7 bp/sec = 1.55 seconds.
- DnaG primase must initiate a new RNA primer on the lagging strand every ~1.5 seconds.

Step 3 (Total Number of Okazaki Fragments):
- Total genome length replicated via lagging strand = half the genome per fork = 2.32 * 10^6 bp.
- Total fragments per fork = 2.32 * 10^6 bp / 1,500 bp = 1,547 fragments.
- Total across both forks = 1,547 * 2 = 3,094 Okazaki fragments synthesized, RNA-excised, gap-filled, and ligated per bacterial cell division.

Conclusion: High-speed replisome coordination requires the Trombone Model: looping of the lagging strand allows the dimeric Pol III core to synthesize both leading and lagging strands simultaneously.

[EXAMPLE 3: Mechanism of Action of Quinolone and Rifamycin Antibiotics]

Problem Context: Ciprofloxacin (a fluoroquinolone) and Rifampicin are frontline antibacterial agents targeting DNA replication and transcription, respectively. Detail their molecular targets, enzyme-drug interaction pockets, and bacterial resistance mutations.

Drug 1 (Ciprofloxacin):
- Molecular Target: Bacterial DNA Gyrase (GyrA subunit) and Topoisomerase IV (ParC subunit).
- Mechanism: Binds the Gyrase-DNA cleavage complex, physically trapping the enzyme in a covalent intermediate where tyrosine residues are covalently linked to 5’-phosphate ends of broken DNA strands. This stabilizes double-strand breaks, colliding with advancing replication forks and causing replisome collapse, SOS response induction, and cell death.
- Resistance Mutation: Point mutations in the Quinoline Resistance-Determining Region (QRDR) of the gyrA gene (e.g., Ser83Leu), reducing drug binding affinity.

Drug 2 (Rifampicin):
- Molecular Target: Beta subunit of bacterial RNA Polymerase (rpoB gene).
- Mechanism: Binds specifically in a pocket adjacent to the catalytic active site within the DNA-RNA channel. It does not inhibit initiation or catalytic phosphodiester bond formation, but sterically blocks elongation once the nascent RNA reaches 2-3 nucleotides in length, preventing promoter clearance.
- Resistance Mutation: Missense mutations in cluster I of rpoB (e.g., Ser531Leu or His526Tyr) that alter the antibiotic pocket without inactivating transcription.

Conclusion: Targeting prokaryotic-specific conformations of replication and transcription enzymes provides high selective toxicity against bacterial pathogens.

[EXAMPLE 4: Mutational Analysis of the -10 and -35 Bacterial Promoter Consensus]

Problem Context: A wild-type E. coli promoter has the sequence: -35 (5'-TTGACA-3'), 17 bp spacer, -10 (5'-TATAAT-3'). Analyze the transcription initiation efficiency changes for the following three laboratory mutations: (A) -10 mutated to 5'-TGTAAT-3', (B) 17 bp spacer mutated to 21 bp spacer, and (C) Up-element insertion (-40 to -60 A-T rich tract).

Mutation A (-10 Hexamer Mutation: TATAAT -> TGTAAT):

- Impact: Severe 'Down-mutation'. The second position A-T pair is critical for forming the initial single-stranded nucleation bubble during open complex formation.
- Outcome: Transcription initiation frequency decreases by >90% because sigma-70 region 2.4 cannot flip aromatic bases out of the duplex into protein pockets, impairing isomerization from closed to open complex.

Mutation B (Spacer Expansion: 17 bp -> 21 bp):

- Impact: Major 'Down-mutation'. In B-form DNA, 10.5 bp corresponds to one 360-degree helical turn. A 4 bp insertion rotates the -35 element relative to the -10 element by ~137 degrees (4 * 34.3 degrees).
- Outcome: The two contact regions of sigma-70 (region 4.2 for -35 and region 2.4 for -10) reside on a rigid protein scaffold. When the elements are twisted to opposite faces of the DNA cylinder, the holoenzyme cannot bind both motifs simultaneously, abolishing promoter recognition.

Mutation C (UP Element Addition at -40 to -60):

- Impact: Powerful 'Up-mutation'. The UP element is an A-T rich sequence recognized directly by the C-terminal domain of the RNA Polymerase alpha subunit (alpha-CTD).
- Outcome: Provides additional high-affinity protein-DNA contacts, increasing promoter binding affinity (KB) up to 30- to 50-fold, characteristic of ultra-strong ribosomal RNA (rrn) promoters.

>> 5.4 University Review & Examination Questions

* Q1: Detail the proofreading exonuclease mechanisms ensuring high fidelity during DNA replication.

Answer: DNA replication exhibits error frequencies of <1 in 10^9 to 10^10 base pairs, achieved through three sequential fidelity stages: (1) Presynthetic nucleotide selection: the active site of DNA Polymerase III geometrically accommodates correct Watson-Crick geometries (error rate ~10^-5). (2) Proofreading: DNA Pol III possesses an intrinsic 3'->5' exonuclease subunit (epsilon subunit). A mispaired 3' terminal base does not fit properly in the polymerase active site; synthesis stalls, the single-stranded primer terminus melts and shifts 3 nm into the exonuclease active site, where the mismatched base is excised hydrolytically (error reduction 10^-2). (3)

Post-replicative Methyl-directed Mismatch Repair (MMR): MutS recognizes mismatches, MutL bridges to MutH which nicks the unmethylated nascent strand at hemimethylated GATC sequences, excising the error (error reduction 10^-3). Combined fidelity = 10^-5 * 10^-2 * 10^-3 = 10^-10.

* Q2: Explain the structural and functional differences between Rho-dependent and Rho-independent transcriptional termination in bacteria.

Answer: Rho-independent (intrinsic) termination relies solely on nucleic acid sequence: a G-C rich dyad symmetry element forms a stable GC-rich RNA hairpin that stalls the moving RNA polymerase, followed immediately by an oligo-uracil tract (7-9 U's). The rU-dA hybrid within the 8-9 bp transcription bubble is the weakest known base-pairing interaction, allowing the transcript to dissociate spontaneously. Rho-dependent termination requires the hexameric ATP-dependent Rho helicase. Rho binds to a C-rich, G-poor rut site on the nascent RNA lacking secondary structure. Rho hydrolyzes ATP to translocate 5'->3' along the transcript, catches the polymerase paused at specific termination pause sites, and uses its RNA-DNA helicase activity to unwind the hybrid, dislodging RNA and polymerase.

>> 5.5 Key Takeaways and Revision Points

* Watson-Crick B-DNA is a right-handed helix (2.0 nm diameter, 10.5 bp/turn, 3.4 nm pitch) with deep major and shallow minor grooves.
* Meselson-Stahl experiment proved semi-conservative replication; leading strand is continuous, lagging strand forms Okazaki fragments.
* Replisome requires Pol III core (alpha, epsilon, theta), beta-clamp (processivity), DnaB (helicase), DnaG (primase), and Gyrase (topoisomerase).
* Bacterial transcription requires Sigma factor for -10 (Pribnow) and -35 promoter recognition; termination is either intrinsic hairpin or Rho-dependent.